Green Screen / Motion Tracking / Compositing

In the past few labs we've been using Adobe After Effects as an animation tool, animating properties of images and videos. In this lab we will be using Adobe After Effects as a compositing tool, or a powerful tool to help us do complex tasks.

In this lab we will be creating a shots for a movie. This movie has many shots that in story take place in beautiful locations around the world. Unfortantely, the movie does not have the budget to fly it's actors and actresses around the world, so instead we will create scenes using After Effects.

Setup

Take a moment to look at the video will will create using Adobe After Effects. You can view the video on YouTube below:

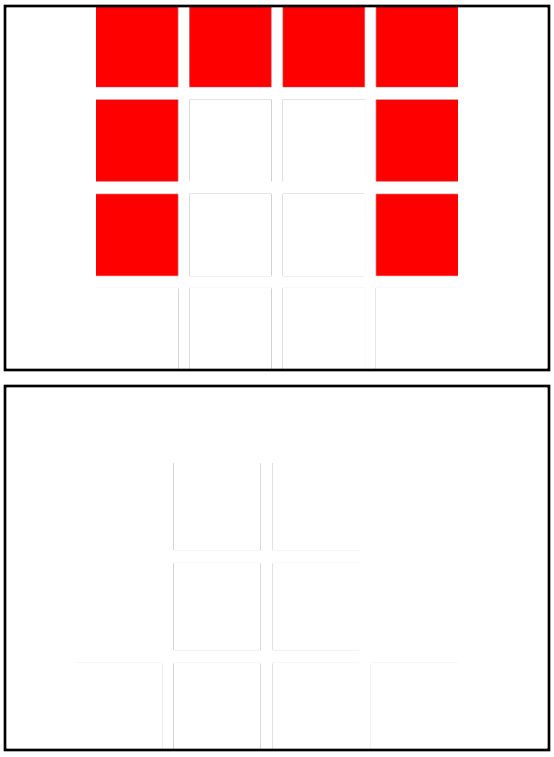
https://www.youtube.com/watch?v=mSI2ig3xSzE

Make sure you have downloaded the lab files needed to complete this lab. The files are in zip folder called *Lab-Post-Production.zip*. Make sure to unzip the folder completely to use the files inside appropiately.

Create a new After Effects project and import all of the files from the lab folder into our project. Create a new composition called *Green Screen that* is *720px x 480px*, frame rate of 30 frames / second and 8 seconds long.

Green Screen Overview

The technique of green screening in film making is when an actor or actress is shot against a green screen. The footage is then taken into a post-production program like Adobe After Effects and the green background is removed. A new background is placed to create the illusion that the actor or actress was filmed in that space.

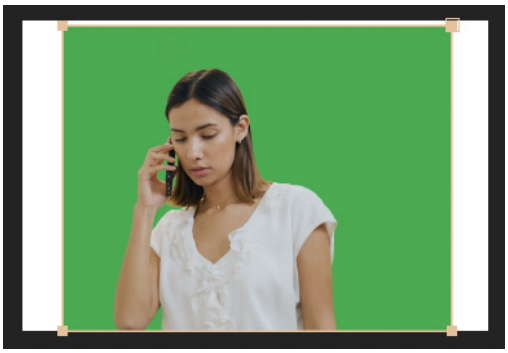


To better understand the concept of green screening take a look at the graphic on the right. You can see a grid of 4x4 squares where some are white and some are red. Imagine that we decided to select all of the red squares and remove them from the graphic we would be left the following image. This is a bit how green screening work, we select green, a color not found typically in human skin, and remove it from the video footage, keeping just the actor or actress.

Part of successfully removing or keying out a green screen

background is for the footage to shot properly to begin with. Below is a link to a video that goes through the process of properly film green screen footage.

Garbage Mask



Drag *Green-Screen-Woman* into the timeline. Scrub through the footage and you'l see a woman talking on the phone. This is the actress in the film. Her character is supposed to be in Paris, but flying actors and actresses to Paris is too expensive, so the director has decided to composite the shot instead. Click on the video *Effiel Tower* and you'll see footage of the Effiel Tower in Paris, France. We will composite the scene so she appears to be there.



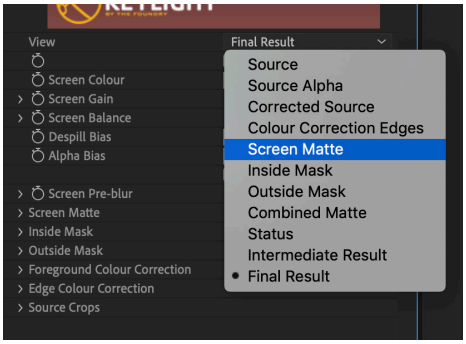
Part of removing a background in After Effects is minimizing the work you need to do. If you view the footage you'll see that we don't need the entire footage, just the footage of the actress. To aid us let's create a layer mask to remove areas of the video we don't need to keep.

You can use a *Rectangular Mask* tool and drag it over the actress, or to get a bit more precision use the *Pen* tool to drag a basic shape around the model. This matte is referred to as garbage matte, because it does not need to precise. Scrub through the footage to make sure that the actress's hands, head and other parts of her body do not get clipped by the matte.

Keying



Select *Green-Screen-Woman* and go to *Effect > Key > Keylight 1.2* to apply a keying affect. Keylight's job is to find and remove a specific color from our footage, in this case we will remove the green. Find the *Screen Colour* option and use the eyedropper to choose a color. Choose the color green just beyond the actress's hair. The area just around the edges of your model is always to hardest to remove. The result will be the background drops out and the actress remains.



Not bad, but we can still make the background removal better. In the *Keylight 1.2* under *View* set it from *Final Result* to *Screen Matte*. The *Screen Matte* will show you what areas were kept and what areas were removed. The kept areas are white and removed areas are black. Sometimes there are areas that gray, these are areas that the Keylight removed some, but not all of the pixels. This is usually do to there being a greenish-tint that occurred during filming.

To make a better cut we want to increase the contrast of the *Screen Matte* to be pure black and white. To help with this under the *Screen Matte* option there is a *Clip Black* and *Clip White*

options. *Clip Black* will take off black colors like dark gray and make them pure black and *Clip White* will take off white colors like light gray and make them pure white. If you view the model you can see that there are some gray areas in her shoulders. To remove this turn the *Clip White* from *100* to *90*. Then set *View* from *Screen Matte* back to *Final Result*.

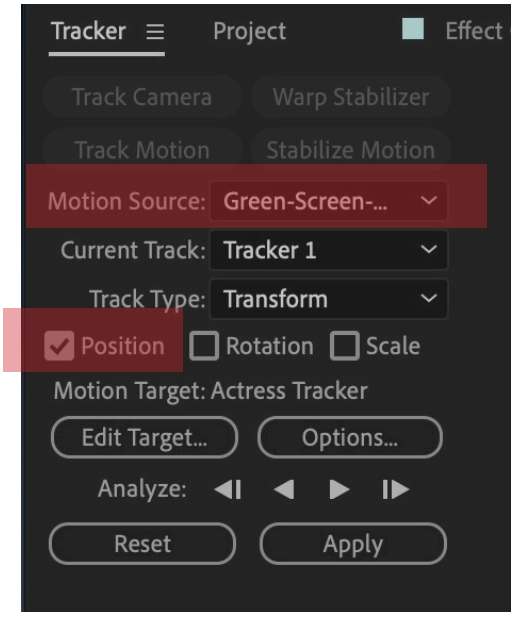


Looks good! To help composite the scene drag the *Effiel Tower* footage and place it in the timeline below the actress footage. Use *Preview* and it will appear like the actress in footage of the Indian temple.



Motion Tracking

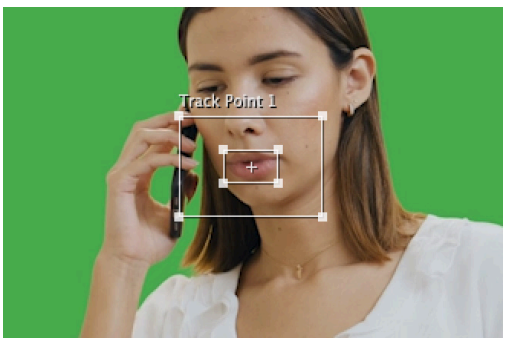
Motion tracking is the idea that we can track points within footage and save that data. That data can then be applied to other objects like text and images.



In the shot of actress on the phone, in the film this is introduction of the character and her name is to appear. The director wants the name text to follow the character as she moves back and forth, to do this we will use motion tracking.

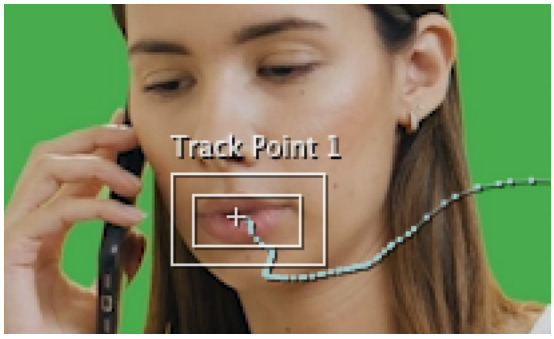
Select *Green-Screen-Woman* and go to *Animation > Track Motion*, this will open up the *Tracker* panel. The *Tracker* panel allows you to setup tracking on a particular video or footage.

In the *Tracker* panel use the *Motion Source* to choose which video we are going to track, in this case it is *Green-Screen-Woman*. Under *Track Type* we want to select *Transform*. Transform allows you to track the position, rotation and scale of an object in a scene. Finally, make sure that just the *Position* checkbox is checked on. At the moment we do not want to use *Rotation* or *Scale*.

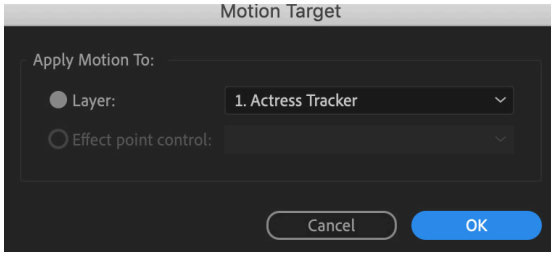


Applying a motion tracker will cause the *Green-Screen-Woman* layer to appear in the *Composition* window and for a track point to appear on the footage. A track point consists of a two squares and a cross hair. You will notice two squares with a crosshair within the scene, this is your track point. The smaller square is the *Feature Region* and should go around an object that you want to track, it helps if this object contrasts from the background like a red dot on a black background. The larger square is the *Search Region*. The *Search Region* is the area that After Effects will track the *Feature Region*. If the object you are tracking jumps around a lot you may want the area to be bigger, if the area tracked does

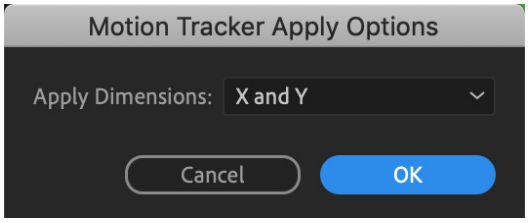
not move much the *Search Region* can be smaller. The larger the *Search Region* the longer it will take After Effects to analyze. The Crosshair represents the point that objects will be attached to. Move the time marker to the beginning of the footage. Use the *Selection* too to place the Feature region around the actresses's mouth. Place the crosshair where the lips separate. In the Tracker panel go to the *Analyze* section and press the *"Play"* button. This will cause After Effects to go through the scene and track the movements of the motion tracker. This process may take a moment or two depending on the power of your computer. At the end you will a motion line of how the object was tracked.



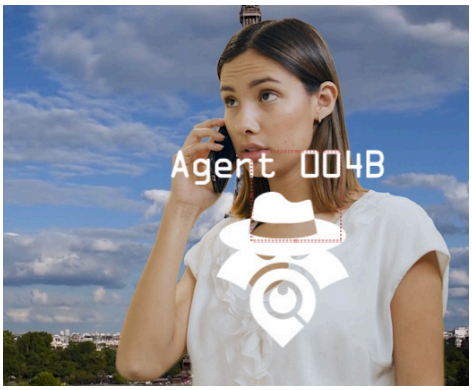
Next let's add spy logo that will be applied on top of the actress drop *Spy-Logo.ai into the timeline* and place it over the actress roughly around the neckline area. When using motion tracker data it is helpful not to apply it directly to a layer, the reason why is we may decide later to offset its position or rotation, that will not be an option if the tracking data is applied directly to the layer. Instead, let's create a null object. The null object will receive the tracking data and then we will parent the logo to the null object. This is helpful because if we decide later to offset the position we can do that to the logo layer itself.



Go to *Layer > New > Null Object* and name it *Actress Tracker*. Select the *Green Screen Woman* and click back on the *Tracker* panel. Under *Edit Target* set *Apply Motion To* to the new *Actress Tracker* null object. This means this is where the tracking data will be saved. A popup menu called *Apply Dimensions* will pop up. This allows to apply X or horizontal data, Y or vertical data or both. Make sure *X and Y* is selected, the hit *OK*.

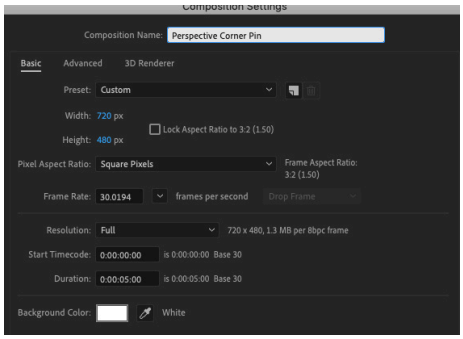


Take a moment to view the *Actress Tracker* and you will see that many position keyframes have been added. Now parent the *Spy-Logo.ai* layer to the *Actress Tracker*. View the results and you'll see that the logo follows as the actress moves back and forth. To offset the logo position use the time marker and place it at the beginning of the timeline. Use the *Spy-Logo.ai* position to offset it just to the right of the actress.



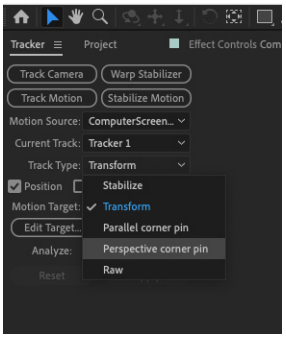
Perspective Corner Pin

Although tracking position is great, sometimes we need more data to do more complicated compositing. The director of the spy movie wants a scene where the a computer is shown running hacker style code. Many times in real production it is hard to film tv and computer screens as they produce glare under production lights, so instead it easier to add the computer content in post.

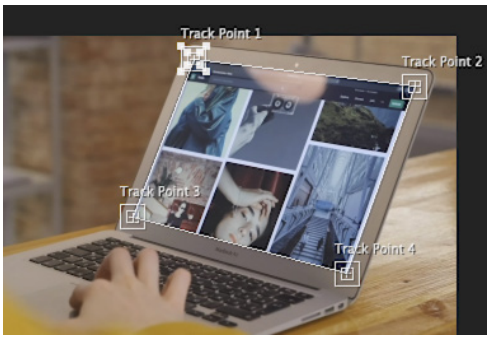


Create a new composition called *Perspectice Corner Pin*

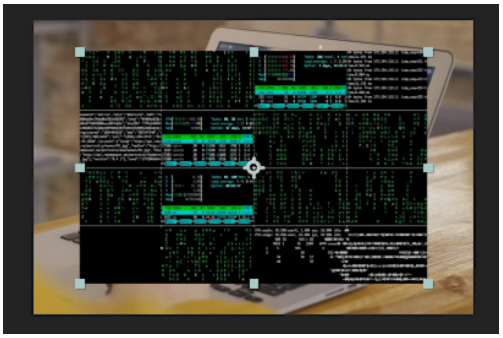
Name: Perspective Corner Pin
Dimensions: 720 x 480px
Frame Rate: 30
Pixel Shape: Square
Duration: 00:00:05:00



Drag the video *ComputerScreen* into the timeline. Scrub through the video and you can see that it is footage of a computer. We would like to replace the screen, but the main problem is that the camera is moving during the shot, this changes the perspective of the screen frame-to-frame. To fix this issue we will use something called *Perspective Corner Pin*.

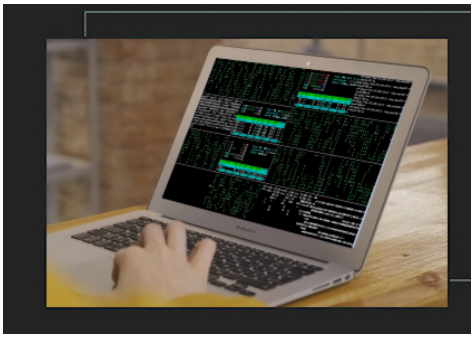


Select the *ComputerScreen* video and go to *Animation > Track Motion* to open up the *Tracker* panel. Set the *Motion Source* to *ComputerScreen* and make sure *Track Type* is set to *Perspective Corner Pin*. This will cause four motion trackers to appear. The idea in order for After Effects to figure out the perspective of the screen it needs to create lines both horizontal and vertical.



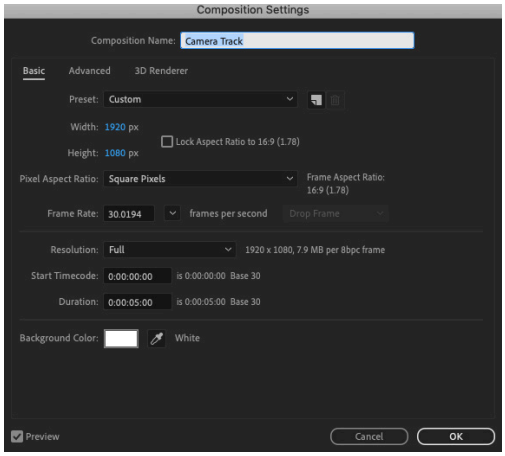
Set the time marker to the beginning of the timeline. Set the motion tracker to the four corners of the computer screen. I recommend using the corners of where the screen meets up with the computer plastic. Make sure the crosshair lines follow the perspective of the screen exactly.

Use the *Analyze* tool to play and track the motion tracker points. Drag the video *Hacker-Code* into the timeline and place it above the *ComputerScreen* layer. Click on the *ComputerScreen* and in the *Tracker* panel set the *Edit Target* to the *Hacker-Code* layer and hit *Apply*. The result will be the *Hacker-Code* footage will be skewed to fit the perspective plane that we tracked. Now it appears as if the computer was looking at the hacker code the whole time!



Camera Tracking

Oh boy, the director really wants something difficult now. They wants us to place our actor into an existing scene. The issue is the shot has the camera turning, meaning our object needs to move in 3D space if its going to look realistic, what are we going to do?



The answer is we are going to use a technique called *Camera Tracking*, what camera tracking does is analyze footage and try to figure out how objects exist in 3D space. This is a bit like the *Perspective Corner Pin*, but even more complex. So let's get started.

Create a new composition called *Camera Track* with the following settings:

Name: Camera Track
Dimensions: 1920 x 720px
Frame Rate: 30
Pixel Shape: Square
Duration: 00:00:05:00

Drag *Video-Turn* into the timeline. Scrub through the footage and you can see that in the footage the camera moves pivoting it's view of the steps and columns. We plan on adding a character to those steps.

Green Screen Man



Create a new compositon called *Green Screen Man* with the following settings:

Name: Green Screen Man
Dimensions: 720 x 480px
Frame Rate: 30
Pixel Shape: Square
Duration: 00:00:05:00

Drag the video *Green-Screen-Man* into the timeline. Scrub through and you can see a man in a suit on his phone. This is going to be the person we composite into the scene.

Use the *Keylight 1.2* effect to key out the green screen background.

Camera Track

Return to the *Camera Track* composition and drag and drop *Video-Turn.mp4* into the composition. You may notice that the video for video-turn is larger dimensions at 1920 x 720. Camera tracking does a better job when a larger, high-quality video is used. It is also important that the video is not resized, thus our composition settings required that the dimensions match the footage.



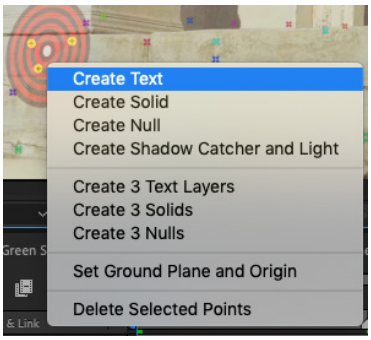
Select the *Video-Turn* layer in the timeline and go to *Animation > Camera Track*. When you do will you'll notice that your footage has a message *"Anaylzing in background"*, this is After Effects analyzing the footage and tracking to find data points it can use later on. Depending on the dimensions of the film and length of the film, this may take a minute or two.

Next go to the *View* menu and make sure that *Show Layer Controls* is turned on. This will allow you to see data points in

your scene. Make sure that the *3D Camera Tracker* effect is highlighted in the *Effects Control* tab. Eventually you will see multicolored croshairs, these are 3D data points. If you scrub the video you can see that as the video moves the data points move in 3D space, awesome!

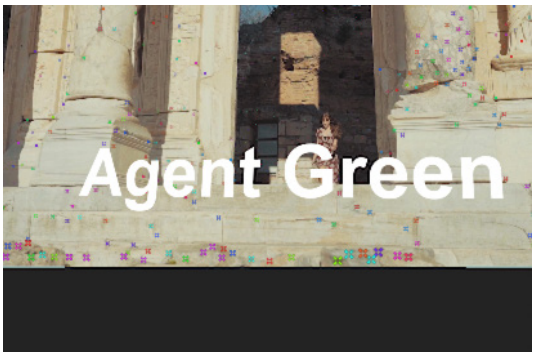


Use the *Selection* tool to highlight a data point, you'll notice it turns yellow to show that it is highlighted. To make sure your object exists in correct 3D space it helps to highlight multiple data points. In this case I clicked and dragged to select three data points just below the column on the left. The result is a bulls-eye appears. Right-click on the bulls-eye and choose the option *"Create Text"*. The result is that a text layer is added to your timeline that is a 3D layer, in addition a camera has also been added to the timeline as well.



Change the text to say *"Agent Green"*. I have the following settings for my text:

Font: Arial Bold
Size: 58

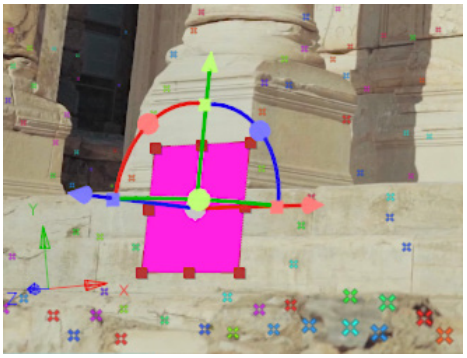


You may notice the the text is tilted downward, you may need to adjust text to properly fit the perspective of the scene. Select the Text layer and use *Z-Rotation* to rotate the text into the proper perspective line. I have mine set to *5˚*.

Scrub through the footage and you can see that the text follows the camera movement in 3D space, awesome!

Add the Man

Select the Video-Turn in the timeline and make sure *3D Camera Tracker* highlight to see the render points again. Choose the same three data points using the *Selection tool*. Right-Click and this time choose *"Create Solid"*. The result will be a colored solid plane is added to your timeline. This solid is where our green-screened actor will go. Once again the 3D perspective is a little off. Use the *Z-Rotation* on the *Track Solid* layer to even the plane to the steps. I set mine to about *353˚*.

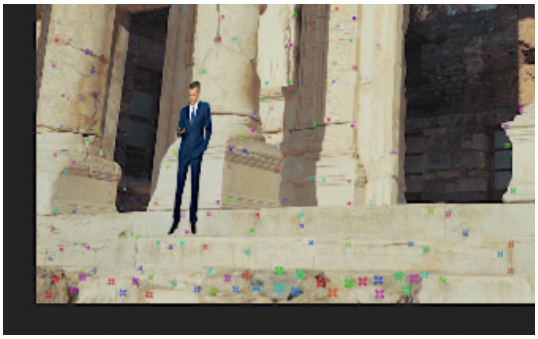


Highlight the *Track Solid* in the timeline. Select the *Green-Screen-Man* composition and hold the *option / alt key* and drag it on top of the *Track Solid* in the timeline. The result will be the composition will replace the *Track Solid* in 3D space. Preview the results and you will see the man spin in 3D space.



The man will be too small for the scene, he looks like an elf. Use the scale to increase his size to better fit the scene. Double check that the text and compositon have the same *Z-Position*. We want

to make sure they exist in the same spot and that one is not in front of the other.



In addition to starting off too small he is also a bit low where we want to position him on the top stair. Use the *Position* property for the *Green Screen Man* composition layer and adjust the y-value to set the man to the top step.

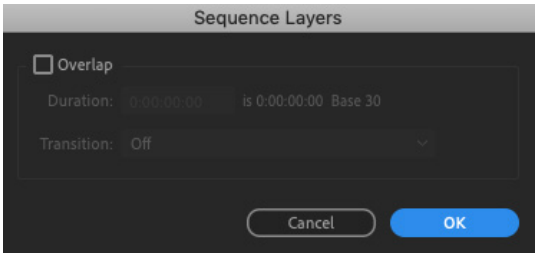
Turn back on the text layer and use the *Preview* panel to view your scene. You should see both the text and green screen composition move in perspective when the camera moves. Great!

Put it All Together

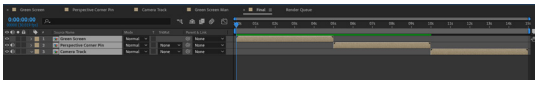
Whew! We've done alot of this film. Let's build a sequence to put all of our work together to show the director.

Build a new composition with the following settings:

Name: Final
Dimensions: 720 x 480px
Frame Rate: 30
Pixel Shape: Square
Duration: 00:00:15:00

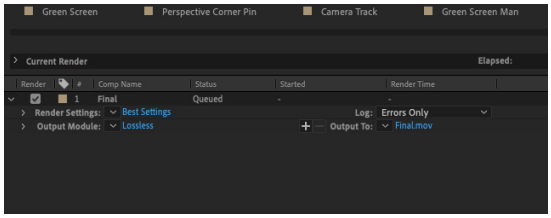


Drag *Green Screen*, *Perspective Corner Pin* and *Camera Track* into the composition. Resize *Camera Track* as that composition had to be larger for the camera track to work properly. Scale it to *45%*.



Let's place each composition next to one another or tip-to-tail as it is referred to. To easily do this select all three composition layers and go to *Animation > Keyframe Assistant > Sequence Layers*, this will automatically line everything up for us. Hit *OK* and you should see your layers laid out in sequential order.

Export



Great job! Now let's export the final project to show the client. With the *Final* composition open in the timeline go to *Composition > Add to Render Queue*.

In the *Render Queue* look for *Comp Name* make sure it is set to *Final*.

Under *Render Settings* make sure it is set to *"Best Settings".*

Under *Output Module* make sure it is set to *"Lossless"* make sure the format is set to either Quicktime or AVI.

Finall, use the *Output To* option save the animation to your Desktop with the label *lastname-post-production.*

Submission

Upload your exported project video to your YouTube account. Publish the video and you will copy and paste the video URL to the submission page for the lab.

In addition to the final video url please upload the *lastname-post-production.aep* file to Canvas for grading. Please do NOT submit all of the files or videos as those are large files, instead just upload the *.aep* project file for grading purposes.